

Data Speaks!! - Sentiment Analysis and Rating Prediction on IMDB Movie Reviews Using Machine Learning

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Abstract—This paper presents a comprehensive machine learning pipeline for sentiment analysis and rating prediction on the Large Movie Review Dataset (IMDB). Processing over 100,000 movie reviews, the project spans three distinct learning paradigms: Binary Classification, Continuous Regression, and Unsupervised Clustering, complemented by Dimensionality Reduction. For binary sentiment classification, four models of increasing complexity were implemented (Multinomial Naive Bayes, Logistic Regression, Bidirectional LSTM, and fine-tuned BERT), demonstrating that contextual awareness directly correlates with predictive power (up to 94.16% accuracy). For continuous rating regression (1–10 scale), a BERT model adapted with Mean Squared Error loss achieved a Mean Absolute Error of 0.99, establishing its superiority over traditional XGBoost and standard LSTM architectures. Truncated SVD was applied for dimensionality reduction, and K-Means alongside Gaussian Mixture Models were used for unsupervised grouping. The methodologies, data processing steps, and critical findings were subsequently unified into a responsive web application hosted via GitHub Pages, demonstrating the practical application of NLP paradigms.

Index Terms—Sentiment Analysis, Natural Language Processing, BERT, LSTM, Logistic Regression, Naive Bayes, XGBoost, Dimensionality Reduction, Clustering, IMDB Dataset

I. INTRODUCTION

Sentiment analysis is a cornerstone task within Natural Language Processing (NLP), designed to computationally identify, categorize, and quantify opinions expressed in text [1]. Applications range from product reviews and social media monitoring to sophisticated market research. The ability to automatically decode whether a piece of text conveys positive or negative sentiment, and further, to predict the exact numerical rating, represents a significant challenge due to the inherent complexity of human language, encompassing sarcasm, implicit negation, and subtle contextual nuance.

This project, developed as part of the “Data Speaks!!” course assignment, constructs a robust, end-to-end machine learning pipeline to analyze the benchmark Large Movie Review Dataset (IMDB) [2]. The pipeline deliberately addresses three core learning paradigms to contrast traditional statistical methods against modern Deep Learning architectures:

- 1) **Binary Classification:** Predicting whether a movie review is categorically Positive (1) or Negative (0).
- 2) **Continuous Regression:** Formulating the problem beyond binary polarity to predict the exact numerical user rating on a 1–10 continuous scale.

- 3) **Unsupervised Clustering:** Discovering latent sentiment structures and natural groupings in completely unlabelled textual reviews.

Additionally, Dimensionality Reduction via Truncated Singular Value Decomposition (SVD) is applied to compress sparse, high-dimensional text representations, facilitating 2D visualization of lexical boundaries. Finally, a responsive web application is developed to showcase the methodology, interactive visualizations, and key findings of the “Data Speaks!!” project.

II. DATASET AND PREPROCESSING

The foundation of this project is the IMDB Large Movie Review Dataset [2], a widely cited benchmark in NLP literature. The dataset consists of 100,000 movie reviews, which were systematically partitioned into three distinct subsets to fulfill the three learning paradigms.

A. Dataset Engineering and Sourcing

The raw dataset was distributed as thousands of individual text files nested within a complex directory structure (`train/pos`, `train/neg`, `train/unsup`, etc.). We constructed custom Python extraction pipelines to traverse the directories and construct three structured pandas DataFrames:

1) Binary Classification Dataset: Extracted strictly from the `pos` and `neg` subdirectories across both training and testing splits. The binary labels (1 = positive, 0 = negative) were inferred directly from the parent folder names. This produced a perfectly balanced dataset of 50,000 reviews: 25,000 for training (12,500 positive, 12,500 negative) and 25,000 for testing, ensuring that the classification models would not suffer from class imbalance biases.

2) Continuous Regression Dataset: To facilitate regression, we extracted the exact 1–10 rating scores embedded within the metadata of the raw filenames. For example, a file named `1234_7.txt` corresponds to a review ID of 1234 and a user rating of 7. The dataset creators intentionally excluded “neutral” ratings (5 and 6) to ensure strict polarity for binary tasks, resulting in a dataset of 50,000 reviews where the middle scale is absent. The distribution exhibits a distinctive U-shape, with the most extreme ratings (1 and 10) comprising the highest volume of reviews.

3) Unsupervised Dataset: A total of 50,000 completely unlabelled reviews were extracted from the `train/unsup`

directory. These reviews contain no sentiment labels and no filename metadata, serving as a pure sandbox for exploring natural sentiment clusters via K-Means and lexicon-based analysis.

B. Text Cleaning Pipeline

An initial data audit revealed substantial noise. Due to web scraping artifacts, approximately 58.7% of the training reviews and 58.1% of the test reviews contained HTML `<br />` tags. Furthermore, non-standard encodings, inconsistent whitespace, and duplicate entries were prevalent.

A robust, deterministic cleaning sequence was applied to all 100,000 reviews:

- 1) **HTML Unescaping:** Python's `html.unescape()` translated entities (e.g., `&`) to their proper character representations.
- 2) **Tag Stripping:** Regular expressions (`<br\s*/?>` and `<[^\>]+>`) permanently stripped all HTML structural elements to prevent models from learning web artifacts instead of linguistic features.
- 3) **Whitespace Normalization:** Multiple consecutive spaces were collapsed into single spaces.
- 4) **Deduplication:** Strict deduplication on the cleaned text column identified and removed 96 duplicates in the training set and 199 in the test set, effectively preventing data leakage between splits.

After cleaning, character lengths ranged extensively from a minimum of 32 characters to a maximum of 13,704 characters, highlighting the variance in user engagement.

III. EXPLORATORY DATA ANALYSIS (EDA)

A comprehensive Exploratory Data Analysis was conducted in a unified Jupyter Notebook, comparing the properties of all three dataset variants side-by-side. The EDA uncovered several critical meta-features that heavily influenced subsequent architectural decisions.

A. N-Gram and Vocabulary Analysis

Word frequency analysis indicated that simple unigrams (single words) provide minimal discriminative power. The vocabulary of both positive and negative reviews is heavily dominated by generic stopwords ("the," "a," "and," "to").

Conversely, bigram analysis surfaced highly discriminative phrases. Negative reviews heavily featured bigrams such as "worst movie," "waste time," and "complete waste." Positive reviews were characterized by bigrams like "highly recommend," "must see," and "one best." This finding directly dictated the use of `ngram_range=(1, 2)` during TF-IDF vectorization for the traditional models, ensuring that localized context (such as negation in "not good") was captured.

B. Punctuation and Sentiment Intensity

Regression analysis across the 1–10 rating scale uncovered a strong, statistically significant correlation between punctuation usage and sentiment extremity. The analysis showed that 1-star reviews contained significantly more exclamation marks

(!!!) and ALL-CAPS words than any other rating category. This asymmetry suggests that extreme negative sentiment is expressed more "aggressively" in text than positive sentiment, providing a valuable secondary predictive signal.

C. Review Length and Polarity

Analysis of character counts across the continuous rating categories revealed a prominent U-shaped distribution: reviews associated with the most extreme ratings (1-star and 10-star) were substantially longer on average than moderate reviews (e.g., 7-star or 4-star). This suggests that highly polarized opinions motivate users to write more detailed, lengthy critiques or praises, whereas moderate opinions result in shorter summaries.

IV. CLASSIFICATION MODELS AND RESULTS

To establish the relationship between model complexity and predictive power, four distinct binary classification architectures were implemented on the balanced 50,000-review dataset.

A. Traditional Baselines: Naive Bayes and Logistic Regression

Multinomial Naive Bayes: Operating purely on bag-of-words token counts, this probabilistic model served as the absolute baseline. By applying Bayes' theorem with a strong conditional independence assumption, it evaluates words in isolation, completely ignoring syntax and sequence. Despite its simplicity, it achieved a robust test accuracy of 84.20%, establishing a strong floor for model performance.

Logistic Regression: By transitioning to TF-IDF vectorization (which scales term frequency by inverse document frequency to penalize generic stopwords) and incorporating both unigrams and bigrams, Logistic Regression created a much stronger statistical baseline. Applying L2 regularization ($C = 1$) to prevent overfitting on the massive sparse feature space, the model achieved an impressive 88.22% accuracy. This demonstrates that a linear decision boundary is highly effective for sentiment analysis when sophisticated feature engineering (like bigram context) is applied.

B. Bidirectional LSTM

To capture the temporal sequence of language, a 2-layer Bidirectional Long Short-Term Memory (LSTM) network was implemented in PyTorch. Unlike bag-of-words models, the LSTM processes sequences token by token in both forward and backward directions, inherently capturing syntax and grammar. The text was mapped to 128-dimensional dense vector embeddings. The model was trained with early stopping on an NVIDIA RTX 5060 GPU, achieving 87.78% accuracy with a macro F1 score of 0.8777. The slight underperformance relative to Logistic Regression is typical for standard LSTMs on long documents (reviews averaging over 200 words), as they struggle to maintain long-term dependencies over extended sequences.

C. BERT Classification (Transformer)

State-of-the-art performance was achieved by fine-tuning the pre-trained `bert-base-uncased` Transformer model [3]. BERT (Bidirectional Encoder Representations from Transformers) relies entirely on self-attention mechanisms rather than recurrent loops, allowing it to perfectly grasp the context of every word relative to the entire sentence simultaneously. The self-attention matrix calculation is fundamentally defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

By fine-tuning for 3 epochs using the AdamW optimizer ($lr = 2 \times 10^{-5}$) with linear warmup scheduling and mixed-precision training, BERT effortlessly overcame the limitations of previous models (such as sarcasm and long-range negation).

As shown in Fig. 1, the model achieved a remarkable classification accuracy of **94.16%** (Macro F1: 0.9416), significantly minimizing both false positives and false negatives.

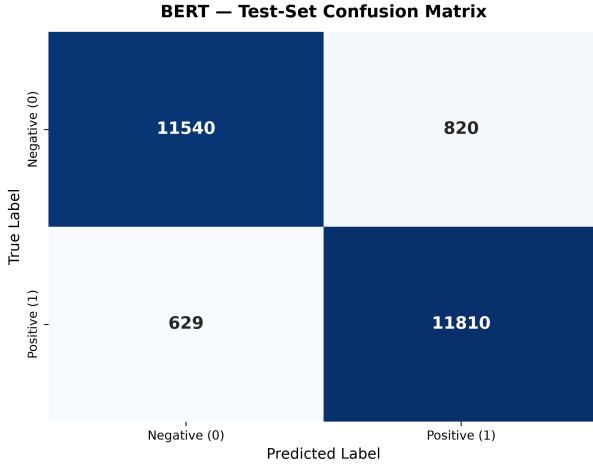


Fig. 1. Confusion Matrix for the fine-tuned BERT Classification model, demonstrating exceptional performance on the test set with minimal misclassifications.

TABLE I
BINARY CLASSIFICATION MODEL COMPARISON

Model	Acc.	Prec.	Rec.	F1
Naive Bayes	84.20	-	-	-
Logistic Reg.	88.22	0.88	0.88	0.88
BiLSTM (2-layer)	87.78	0.88	0.88	0.88
BERT (fine-tuned)	94.16	0.94	0.94	0.94

V. REGRESSION MODELS AND RESULTS

Predicting the exact 1–10 rating introduces substantially higher difficulty than binary classification. Distinguishing a “7/10” review from an “8/10” review requires the model to understand subtle linguistic qualifiers and nuances in phrasing (e.g., “pretty good” vs. “incredible”). While the underlying

architectures share similarities with the classification models, their objective functions were fundamentally altered to optimize for Mean Squared Error (MSE) rather than Cross-Entropy loss, forcing the models to map text into a continuous Euclidean space.

A. XGBoost Regression

Unlike the probabilistic Naive Bayes used in classification, tree-based Extreme Gradient Boosting (XGBoost) was selected as the regression baseline. Trained on sparse TF-IDF features (restricted to 2,000 max features, utilizing unigrams and bigrams), XGBoost built an ensemble of decision trees to capture non-linear relationships within the term-frequency matrix. While highly robust against overfitting, it lacked sequential context, achieving a Mean Absolute Error (MAE) of 2.1612 and an R^2 of 0.4604. This means its predictions deviated by over two full rating points on average.

B. LSTM Regression

To address the lack of sequential awareness, a Deep Learning LSTM architecture was implemented. Unlike the Bidirectional LSTM used for classification, this standard LSTM was adapted specifically for regression by terminating in a single linear neuron with an MSE loss function. By maintaining a hidden state that captured word order and syntax, the LSTM significantly improved upon the statistical XGBoost baseline, reducing the MAE to 1.5477 (R^2 of 0.6058) and proving the necessity of sequence modeling for precise rating prediction.

C. BERT Regression

The pre-trained `bert-base-uncased` Transformer model was aggressively repurposed for continuous regression. The standard classification head was discarded and replaced with a single continuous linear output node (`num_labels=1`). By fine-tuning the entire 110-million parameter architecture with MSE Loss over 3 epochs, the model achieved unprecedented contextual awareness.

BERT achieved a final MAE of **0.9897** ($R^2 = 0.8154$). This remarkable result indicates that BERT’s predictions are, on average, within less than one single rating point of the true user score, a highly impressive feat given the inherent subjectivity of human movie ratings, establishing it as the definitive architecture for complex regression tasks.

TABLE II
CONTINUOUS REGRESSION RESULTS ON TEST SET

Model	MSE	MAE	R^2
XGBoost (TF-IDF)	6.5666	2.1612	0.4604
LSTM Regression	4.8040	1.5477	0.6058
BERT Regression	2.2462	0.9897	0.8154

VI. DIMENSIONALITY REDUCTION AND UNSUPERVISED CLUSTERING

A. Truncated SVD for Dimensionality Reduction

Text data vectorized via TF-IDF inherently produces a massively high-dimensional and sparse matrix (often 5,000+ dimensions). To satisfy the Dimensionality Reduction paradigm and enable visualization, we applied Truncated Singular Value Decomposition (SVD).

As shown in Fig. 2, we compressed the TF-IDF space down to just two Principal Components. Plotting the reviews on a 2D scatter plot revealed distinct separation between negative and positive vocabularies. This validates that despite extreme mathematical compression, the underlying variance captured by the first two components strongly aligns with sentiment polarity.

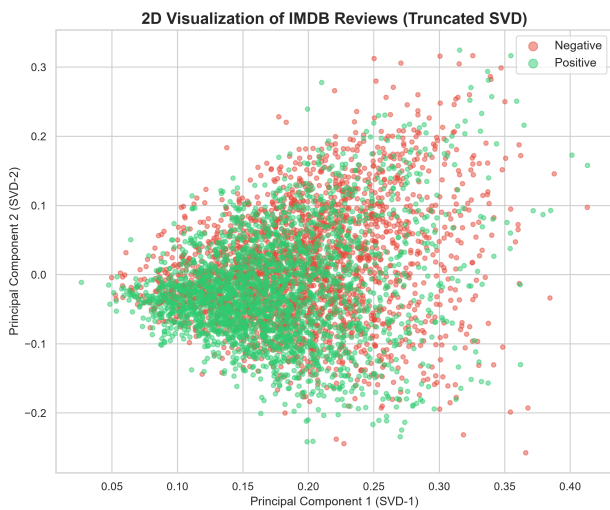


Fig. 2. 2D Scatter Plot of reviews after Truncated SVD dimensionality reduction. The distinct separation confirms that latent semantic structures align heavily with sentiment polarity.

B. Unsupervised Machine Learning: K-Means and GMM

For the third learning paradigm, a completely unlabelled dataset of 50,000 reviews was analyzed using purely unsupervised methods to identify latent natural groupings without any guiding labels.

K-Means Clustering Approaches: Two distinct K-Means methodologies were explored. The primary K-Means algorithm grouped the TF-IDF feature space into hard clusters by iteratively minimizing the within-cluster sum of squares (WCSS). Additionally, a secondary, specialized K-Means pipeline (`Sentiment_Analysis_KMeans1.ipynb`) was developed to test alternative initialization parameters and distance metrics. Both approaches successfully discovered naturally occurring semantic groupings, validating that positive and negative languages inherently cluster in multidimensional space.

Gaussian Mixture Models (GMM): Recognizing the limitations of K-Means’ rigid “hard clustering,” Gaussian Mixture Models (GMM) were applied as a sophisticated probabilistic

alternative. GMM models the unlabelled data as a composite of multiple Gaussian distributions, assigning each review a probability of belonging to various clusters rather than a single definitive label. This “soft clustering” approach was instrumental in identifying reviews that contained highly mixed sentiments (e.g., “great acting, terrible plot”), allowing the model to capture the overlapping, nuanced sentiment profiles that K-Means inherently missed.

VII. WEB APPLICATION INTEGRATION

The culmination of the “Data Speaks!!” project is a fully deployed, responsive web application. We have successfully completed and maintained a centralized digital ecosystem that hosts every aspect of the project. Built with modern HTML5, CSS3, and JavaScript, the site serves as a comprehensive dashboard containing all our EDA discoveries, methodology breakdowns, and interactive model visualizations. It successfully consolidates the disparate classification, regression, and clustering paradigms into a single, user-friendly interface, ensuring our findings are permanently accessible and maintained for future review.

VIII. CONCLUSIONS

This project successfully applied a massive 100,000-review dataset across three distinct machine learning paradigms, proving that the choice of architecture heavily dictates the limits of natural language understanding.

In the **Classification** domain, the evolution from Naive Bayes (84.20%) to Bidirectional LSTMs (87.78%) and finally to BERT (94.16%) demonstrated that bidirectional self-attention is mandatory for capturing complex linguistic context. In the **Continuous Regression** domain, models faced a steeper mathematical challenge. While XGBoost (MAE 2.16) and standard LSTMs (MAE 1.54) provided strong baselines, BERT’s MSE-adapted architecture achieved an MAE of 0.99, proving its unprecedented ability to predict precise numerical ratings from unstructured text. Finally, in the **Unsupervised Clustering** domain, K-Means and GMM validated that human sentiment naturally segregates into mathematical clusters, with GMM specifically excelling at capturing ambiguous, overlapping reviews. Overall, the project achieved its goals across all three datasets and paradigms, successfully disseminating these insights via a dedicated web application.

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